Day 16 Study Notes

Sat 28 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Wave optics (part 2: diffraction and polarisation) (75 min) · Maths: Functions (part 1) (75 min) · Chemistry: Solutions (part 1) (60 min)
Exam	1 h 15 min	The Day 16 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Wave optics (part 2: diffraction and polarisation)

This finishes wave optics: diffraction at a single slit, resolving power and polarisation. Brewster's law and Malus's law are the usual numericals.

1. Diffraction at a single slit

Slit width a, screen distance D, wavelength λ.

- Minima where $a \sin\theta = n\lambda$ ($n = 1, 2, \ldots$).
- Secondary maxima roughly where $a \sin\theta = \left(n + \frac{1}{2}\right)\lambda$, each much fainter than the central one.
- The central maximum spans from $-\frac{\lambda}{a}$ to $+\frac{\lambda}{a}$: angular width $\frac{2\lambda}{a}$, linear width $\frac{2\lambda D}{a}$. It is twice as wide as each secondary maximum.
- A narrower slit or a longer wavelength gives a wider pattern.
- Diffraction is noticeable when the opening or obstacle is comparable to the wavelength.
- **Fresnel distance** $z_F = \frac{a^2}{\lambda}$: up to this distance ray optics is a good approximation.

2. Interference and diffraction compared

Interference (double slit)	Diffraction (single slit)
Two coherent sources	Wavelets from different parts of one opening
Fringes of equal width	Central maximum twice as wide as the rest
All bright fringes about equally bright	Intensity falls quickly away from the centre

3. Resolving power

- Telescope: smallest resolvable angle $\theta = \frac{1.22\lambda}{D}$, where D is the objective's diameter. A larger objective resolves finer detail.
- Microscope: resolving power rises with the numerical aperture and with shorter wavelength (the reason electron microscopes resolve more).

4. Polarisation

- Polarisation shows that light is a **transverse** wave. Sound, being longitudinal, cannot be polarised.
- Unpolarised light through one polaroid: intensity halves, $I = \frac{I_0}{2}$.
- **Malus's law:** polarised light of intensity I_1 through an analyser at angle θ to its axis gives $I = I_1 \cos^2\theta$.
- **Brewster's law:** at the polarising angle θ_p, $\tan\theta_p = n$. The reflected light is completely polarised, and the reflected and refracted rays are at 90° to each other. For glass ($n = 1.5$), $\theta_p \approx 57^\circ$.
- Scattered sunlight seen at 90° to the sun's direction is polarised.

Solved example 1. Light of wavelength 500 nm falls on a slit 0.1 mm wide; the screen is 1 m away. Find the width of the central maximum.

$$\frac{2\lambda D}{a} = \frac{2 \times 5 \times 10^{-7} \times 1}{10^{-4}} = 10^{-2}\ \text{m} = 1\ \text{cm}.$$

Solved example 2. Unpolarised light of intensity I_0 passes through two polaroids whose axes are at 60°. Find the final intensity.

After the first: $\frac{I_0}{2}$. After the second: $\frac{I_0}{2}\cos^2 60^\circ = \frac{I_0}{8}$.

Traps. The first polaroid halves unpolarised light; Malus's law applies only after that. At Brewster's angle, use $\tan$, not $\sin$. The central diffraction maximum has width $\frac{2\lambda D}{a}$, twice $\frac{\lambda D}{a}$. Slit width a in diffraction is not slit separation d in interference.

Mathematics · Functions (part 1)

Functions give one to two questions per paper, usually domain and range, or counting one-one and onto functions. Today covers those. Part 2 covers composition, inverse and periodic functions.

1. Domain: the rules

- Under a square root (or any even root): the expression must be ≥ 0.
- In a denominator: the expression must be $\neq 0$. Under a root in a denominator: > 0.
- Inside $\log_a$: the argument must be > 0 (and the base $a > 0$, $a \neq 1$).
- Inside $\sin^{-1}$ or $\cos^{-1}$: between -1 and 1.
- For a sum or product of functions, the domain is the intersection of the separate domains.

2. Range: methods

- Write $y = f(x)$, solve for x in terms of y, and find the y for which that x is real and in the domain.
- If this gives a quadratic in x, require its discriminant ≥ 0.
- Use known bounds: $x^2 \geq 0$, $-1 \leq \sin x \leq 1$, $a^x > 0$, AM–GM.
- Example: $f(x) = \dfrac{x}{1+x^2}$. $yx^2 - x + y = 0$ has real x when $1 - 4y^2 \geq 0$, so the range is $\left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$.

3. Standard functions

Function	Domain	Range
$\lvert x \rvert$	$\mathbb{R}$	$[0, \infty)$
$[x]$ (greatest integer)	$\mathbb{R}$	integers
$\{x\} = x - [x]$ (fractional part)	$\mathbb{R}$	$[0, 1)$
$\operatorname{sgn} x$	$\mathbb{R}$	$\{-1, 0, 1\}$
e^x	$\mathbb{R}$	$(0, \infty)$
$\ln x$	$(0, \infty)$	$\mathbb{R}$

$[x]$ is the greatest integer not exceeding x: $[3.7] = 3$, $[-2.5] = -3$, $[-0.3] = -1$.

4. Types of functions

- **One-one (injective):** different inputs give different outputs. Test: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; graphically, every horizontal line cuts the graph at most once. A strictly increasing or decreasing function is one-one.
- **Onto (surjective):** the range equals the codomain.
- **Bijective:** both. Only bijections have inverses.
- $f(x) = x^2$ from $\mathbb{R}$ to $\mathbb{R}$ is neither one-one ($f(1) = f(-1)$) nor onto (no negative outputs).

5. Counting functions from a set of m elements to a set of n elements

- All functions: n^m.
- One-one functions ($m \leq n$): ${}^nP_m = n(n-1)\cdots(n-m+1)$.

- Onto functions to a set of 2 elements: $2^m - 2$. In general, by inclusion–exclusion: $n^m - \binom{n}{1}(n-1)^m + \binom{n}{2}(n-2)^m - \dots$
- Bijections ($m = n$): $n!$.

6. Even and odd functions

- Even: $f(-x) = f(x)$, graph symmetric about the y-axis (e.g. x^2, $\cos x$, $|x|$).
- Odd: $f(-x) = -f(x)$, graph symmetric about the origin (e.g. x^3, $\sin x$, $\tan x$).
- Every function on a symmetric domain is the sum of an even part $\dfrac{f(x) + f(-x)}{2}$ and an odd part $\dfrac{f(x) - f(-x)}{2}$.

Solved example 1. Find the domain of $\dfrac{1}{\sqrt{x^2 - 5x + 6}}$.
Need $x^2 - 5x + 6 > 0$, that is $(x-2)(x-3) > 0$: $x < 2$ or $x > 3$.

Solved example 2. How many onto functions are there from $\{1, 2, 3\}$ to $\{a, b\}$?
All functions: $2^3 = 8$. Remove the 2 constant functions: 6.

Traps. A root in the denominator needs strictly positive. $\lceil -2.5 \rceil = -3$, not -2. The range of x^2 on $\mathbb{R}$ is $[0, \infty)$, so it is not onto $\mathbb{R}$. Counting functions: the number of choices per element is the size of the codomain, raised to the size of the domain.

Chemistry · Solutions (part 1)

Solutions gives one to two questions per paper. Today covers concentration terms, Henry's law, Raoult's law and deviations from ideality. Part 2 covers colligative properties.

1. Concentration terms

Term	Definition	Changes with temperature?
Mass percentage	$\dfrac{\text{mass of solute}}{\text{mass of solution}} \times 100$	No
Mole fraction x	$\dfrac{\text{moles of component}}{\text{total moles}}$; all fractions add to 1	No
Molality m	moles of solute per kg of **solvent**	No
Molarity M	moles of solute per litre of **solution**	Yes (volume changes)
ppm	parts of solute per million parts of solution	No (by mass)

- Molarity from molality: $m = \dfrac{1000M}{1000d - M \cdot M_2}$, where d is the density in g/mL and M_2 the molar mass of the solute.

2. Henry's law

- The partial pressure of a gas over a solution is proportional to its mole fraction in the solution: $p = K_H x$.
- A larger K_H means a **less** soluble gas at the same pressure.
- Gas solubility falls as temperature rises (K_H increases).
- Uses: soft drinks are bottled under high CO_2 pressure; divers breathing compressed air risk the bends, so tanks use helium-diluted air; at high altitude low oxygen pressure causes anoxia.

3. Raoult's law

- For two volatile liquids: $p_A = p_A^\circ x_A$, $p_B = p_B^\circ x_B$, total $P = p_A^\circ x_A + p_B^\circ x_B$.
- Composition of the vapour: $y_A = \dfrac{p_A}{P}$. The vapour is richer in the more volatile component.
- Non-volatile solute: $P = p^\circ_{solvent}\, x_{solvent}$, so the vapour pressure falls.

4. Ideal and non-ideal solutions

	Ideal	Positive deviation	Negative deviation
A–B attraction compared with A–A and B–B	same	weaker	stronger
Vapour pressure	follows Raoult	higher than Raoult	lower than Raoult
ΔH_{mix}, ΔV_{mix}	0, 0	positive	negative
Examples	benzene + toluene; n-hexane + n-heptane	ethanol + water; ethanol + acetone; acetone + CS_2	chloroform + acetone; HNO_3 + water; phenol + aniline
Azeotrope	none	minimum boiling (ethanol–water, 95% ethanol)	maximum boiling (HNO_3–water, 68% HNO_3)

- An azeotrope boils at constant temperature with vapour of the same composition as the liquid, so it cannot be separated by fractional distillation.
- For an ideal solution $\Delta S_{mix} > 0$ and $\Delta G_{mix} < 0$: mixing is still spontaneous.

Solved example 1. Find the molarity of 4 g of NaOH in 250 mL of solution.
$\frac{4}{40} = 0.1$ mol in 0.25 L: 0.4 M.

Solved example 2. Liquids A ($p^\circ = 100$ mm Hg) and B ($p^\circ = 60$ mm Hg) form an ideal solution with $x_A = 0.25$. Find the total vapour pressure.
$P = 100 \times 0.25 + 60 \times 0.75 = 25 + 45 = 70$ mm Hg.

Traps. Molality uses kg of solvent, molarity litres of solution. Higher K_H means lower solubility. Ethanol–water shows positive deviation; chloroform–acetone shows negative deviation (hydrogen bonding between them). An ideal solution still has $\Delta S_{mix} > 0$.